THE CELL
GENERAL STRUCTURE & FUNCTION

BIOLOGY course _ LEC.2

UZ. Alaa Moawia Ahmed

Cells

- Smallest living unit
- Most are microscopic

DISCOVERY OF CELLS

- Robert Hooke “English Father of Microscopy”
  - Observed sliver of cork
  - Saw “row of empty boxes”
  - Coined the term cell

THE CELL THEORY

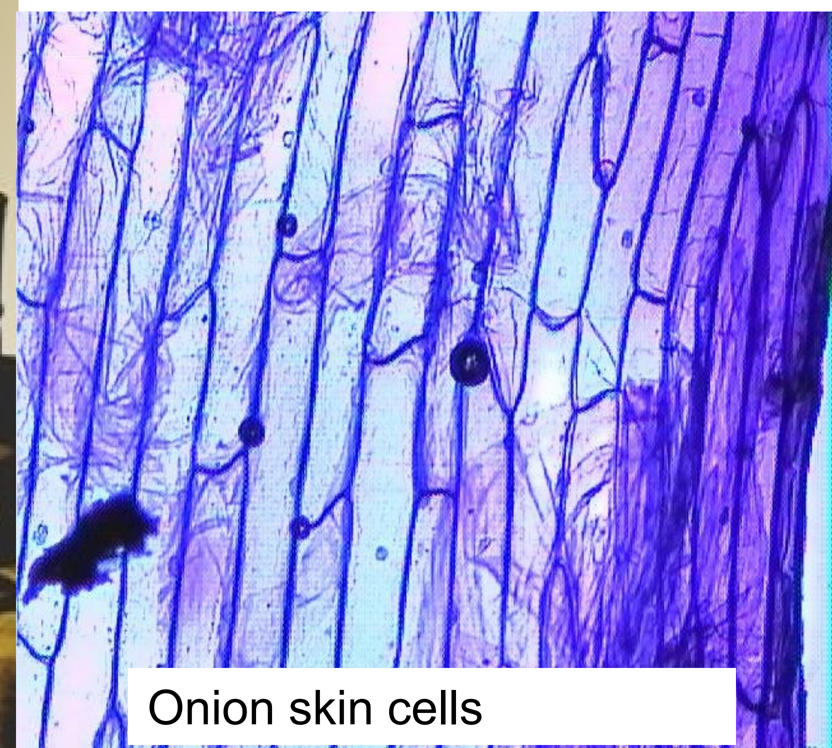
The cell theory is made up of these three facts:

1. All living things are made up of cells & the products of those cells.
2. All cells carry out their own life functions.
3. New cells come from other living cells.

Who came up with this theory?

1. **Schleiden (circa 1838)**

All plants are made of cells!

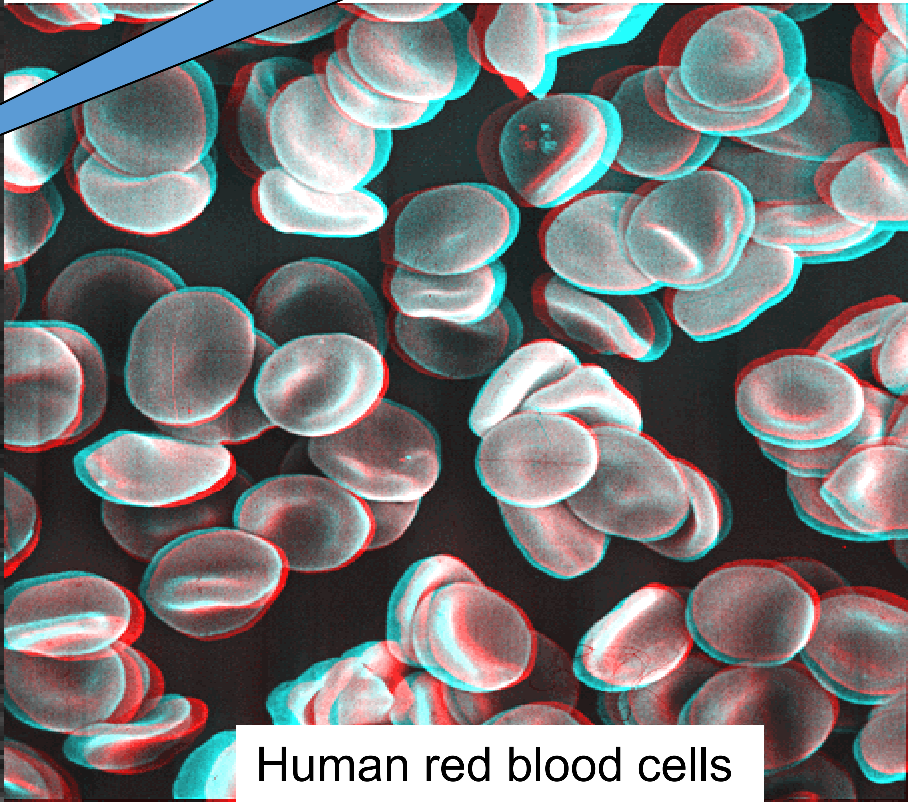


Onion skin cells

Who came up with this theory?

2. Schwann

All animals are made of cells!

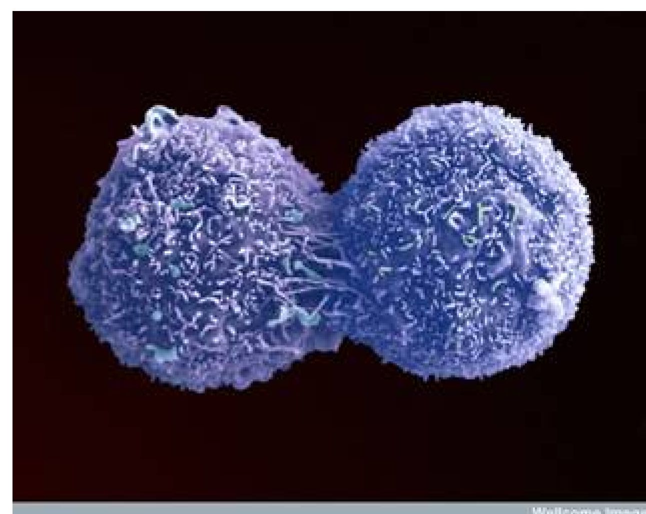


Human red blood cells

Who came up with this theory?

3. Virchow

All cells come from pre-existing cells!



Exceptions to the Cell Theory

1. **Viruses:**

   According to the Cell Theory: The virus is not a cell. Viruses are made of two chemicals, protein & nucleic acid. They have no membranes, nucleus, or protoplasm. They appear to be alive when they reproduce after infecting a host cell.

2. **Mitochondria & chloroplasts:**

   These cell organelles (small structures inside the cell) have their own genetic material & reproduce independently from the rest of the cell.

THE CELL

- The cell is the basic structural and functional unit of all known living organisms.

- A cell is the smallest living unit of organisms.

- The largest known cell is that found in the ostrich egg.

- Some organisms, such as most bacteria, are unicellular (consist of a single cell). Other organisms, such as humans, are multicellular. (Humans have an estimated 100 trillion cell).

- The cell is made up of a mass of protein material called “protoplasm”.

- On closer examination the protoplasm is found to be made of many different substances in different shapes.

- Cells are usually so small, they can only be seen using a microscope.

- The human body contains many different organs, such as the heart, lung, and kidney, with each organ performing a different function, Also cell have a set of "little organs," called organelles, that are specialized for carrying out one or more important functions.

- Most organelles can only be seen with a so-called “electron Microscope.

Component of the cell

The three Major component of the animal cell:-

- **Cytoplasm**
- **Plasma membrane**
- **Nucleus**

Prokaryotes Vs. Eukaryotes

- All living things can be categorized as being prokaryotic or eukaryotic.

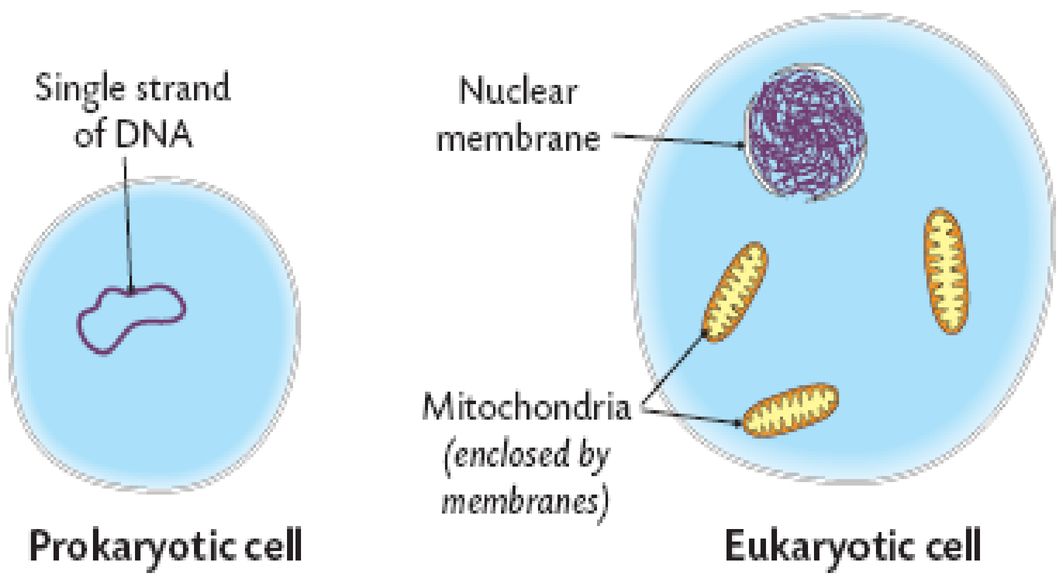
- **Prokaryotes:** no nucleus and no membrane-bound organelles (nuclei, mitochondria, or chloroplasts.
- All prokaryotes are placed in the Kingdom Monera i.e. the bacteria.

- **Eukaryotic cells** have a

1.Membrane-bound nucleus.

2.Membrane-bound organelles such as nuclei,

3.Mitochondria and chloroplasts are only present in eukaryotic cells. The Protista, Fungi, Plants and Animals are eukaryotic organisms.



THE CYTOPLASM

- The term cytoplasm or cytosome is applied to all protoplasm lying outside the nucleus.

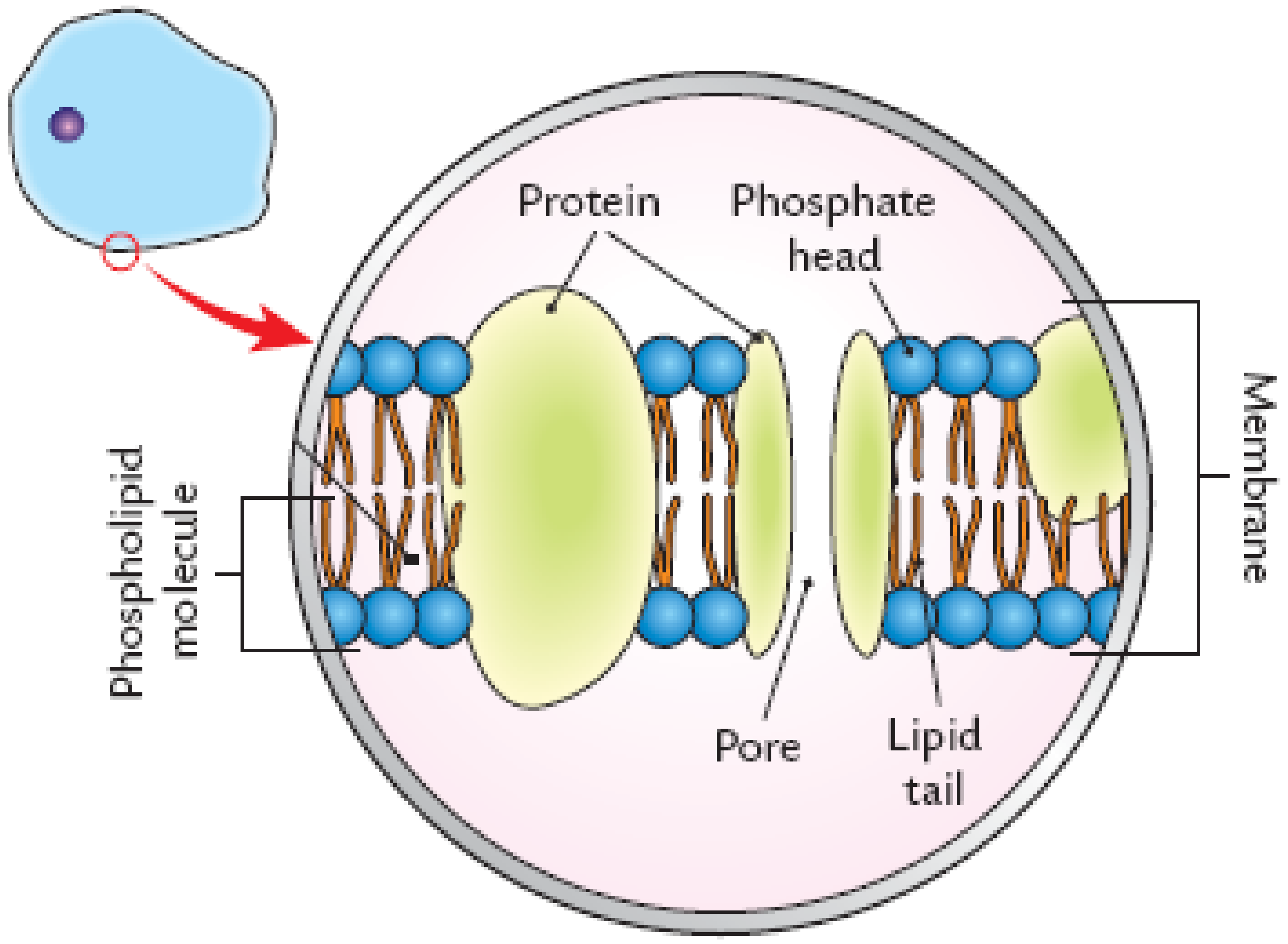
- It is limited externally by the plasma membrane, which is the outer living part of the protoplasm and is usually invisible.

THE PLASMA MEMBRANE

- The plasma membrane is a thin membrane which plays an important role in the exchange of materials which occur between the cell and it surround.

- This membrane serves to separate and protect a cell from its surrounding environment and is made mostly from a double layer of lipids .

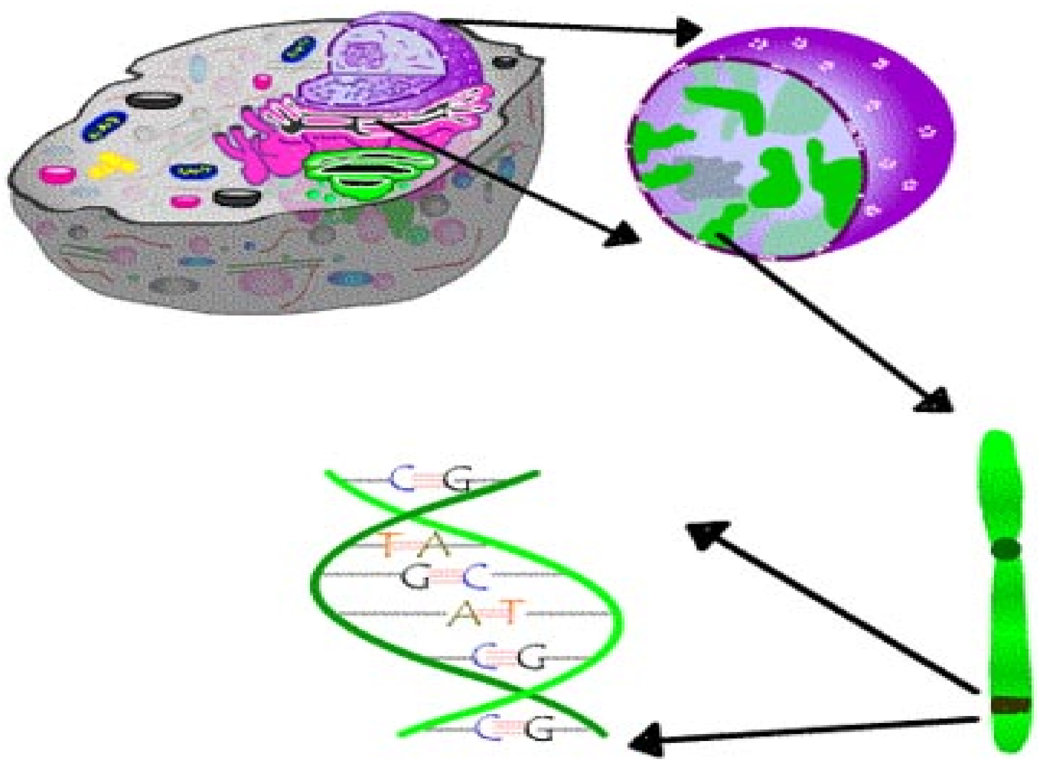
- In plants and prokaryotes is usually covered by a cell wall.



Protein
Phosphate head
Membrane
Phospholipid molecule
Pore
Lipid tail

THE NUCLEUS

- It contains **GENETIC MATERIAL**, on which the DNA consisting of genes the that codes for proteins, the main ingredients of living beings.



- Nucleus is a highly specialized organelle that serves as the information processing and administrative center of the cell.

- This organelle **has two major functions:**

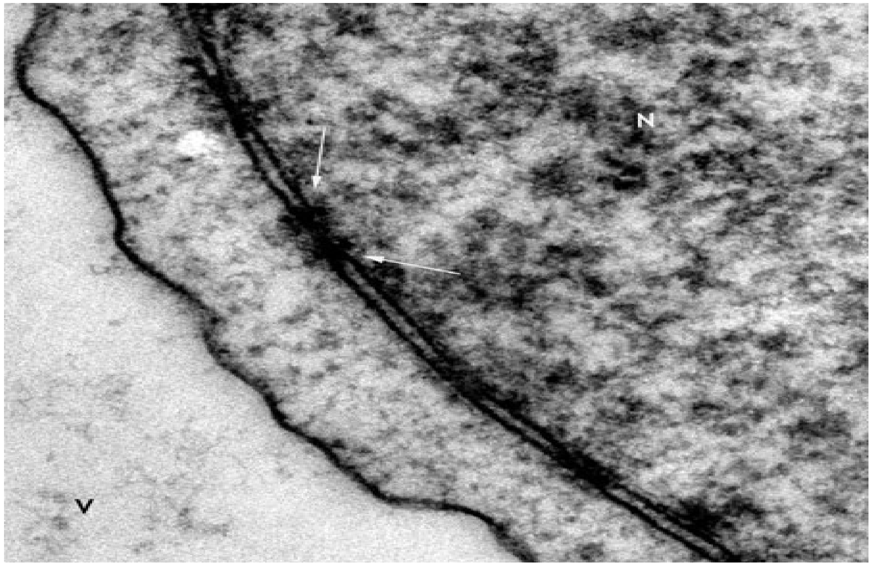
1. It stores the cell's hereditary material, or DNA, and
2. It coordinates the cell's activities, which include growth, protein synthesis, and reproduction (cell division)…

- The nucleus is surrounded by the nuclear membrane.

The Nuclear Membrane & nucleus

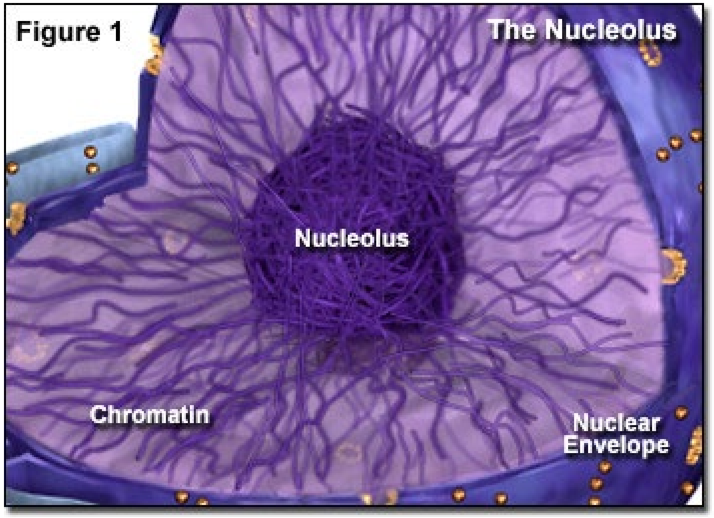
- It surround nucleus & separates the nuclear mass from the rest of the cell. It has many openings (**PORES**).

- Materials can enter and leave the nucleus: e.g. **RNA** from nucleus to cytoplasm and **Nucleotides** from cytoplasm to nucleus.



THE NUCLEOLUS

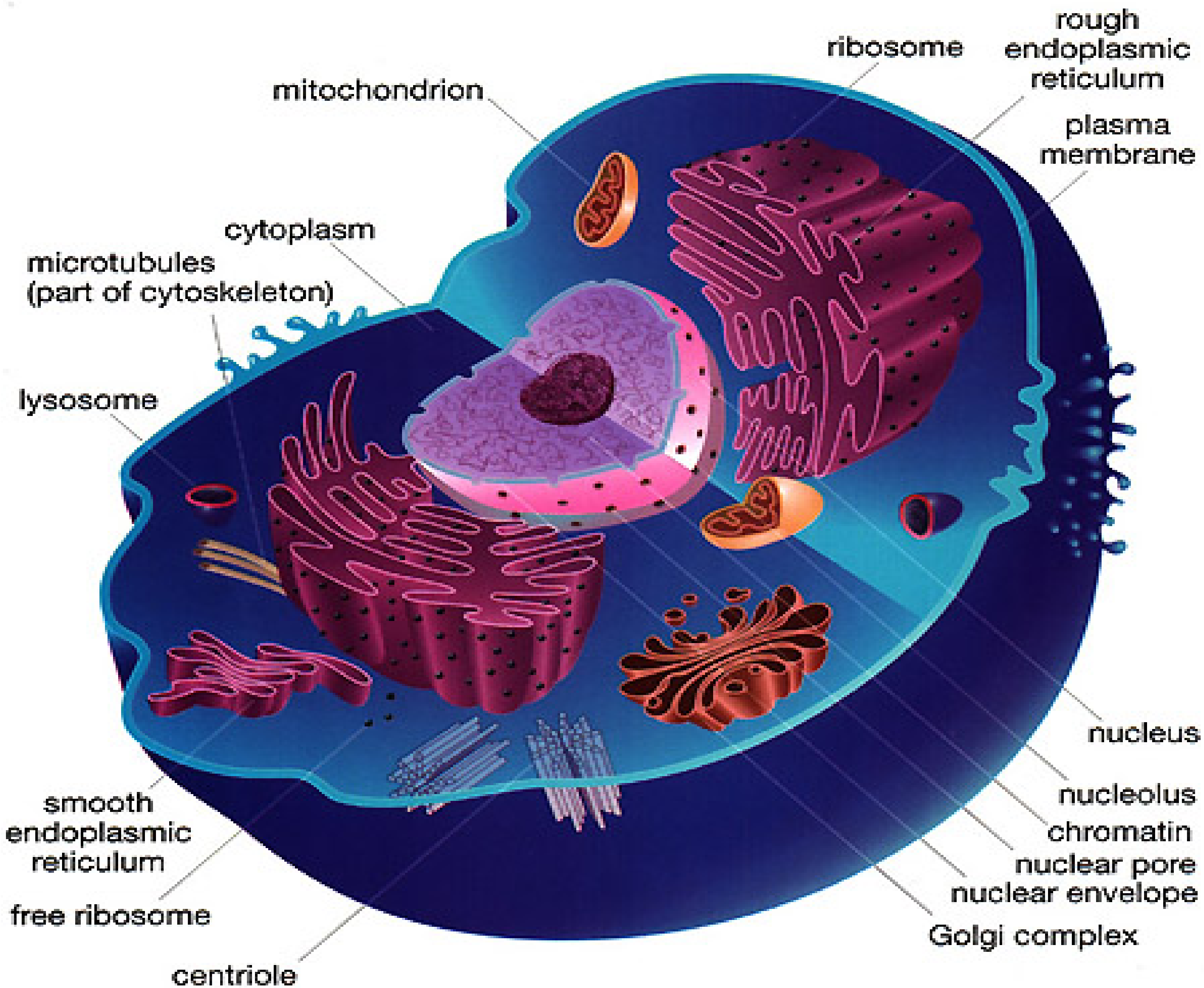
- Is the spherical shape area in the center of the nucleus where ribosomes are made from rRNA (ribosomal RNA).



THE GENETIC MATERIAL

- Two different kinds of genetic materials exist: Deoxyribonucleic acid (DNA) and Ribonucleic acid (RNA).

- DNA the hereditary material which made up of genes, and RNA containing the information necessary to build various proteins such as enzymes.



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ENDOPLASMIC RETICULUM

- The endoplasmic reticulum is a network of sacs that manufactures, processes, and transports chemical compounds for use inside and outside of the cell.

- It is connected to the double-layered nuclear envelope, providing a pipeline between the nucleus and the cytoplasm.

- Provides a large surface area for organization of chemicals reactions and synthesis.

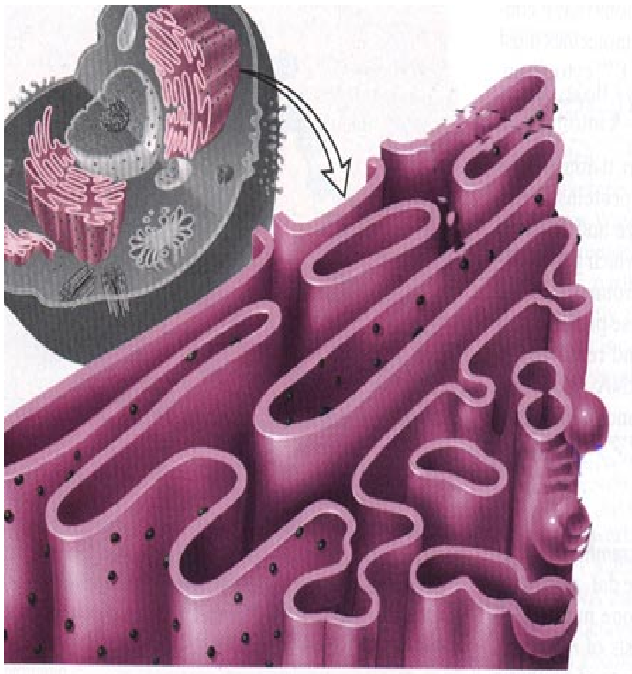
There are two types of (ER):

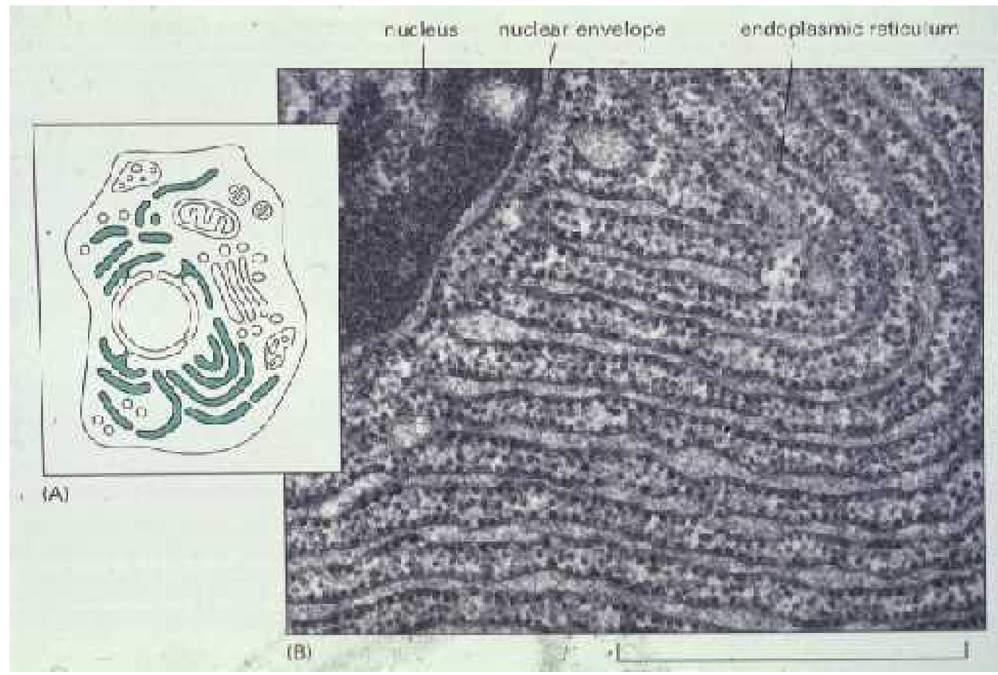
- **Rough (ER):** with ribosomes attached to its surface. Its functions: protein synthesis and transportation.

- **Smooth (ER):** without ribosomes.

  Its functions: lipid and steroids synthesis, it may have other function e.g. in the liver cell it helps in detoxifying drugs.

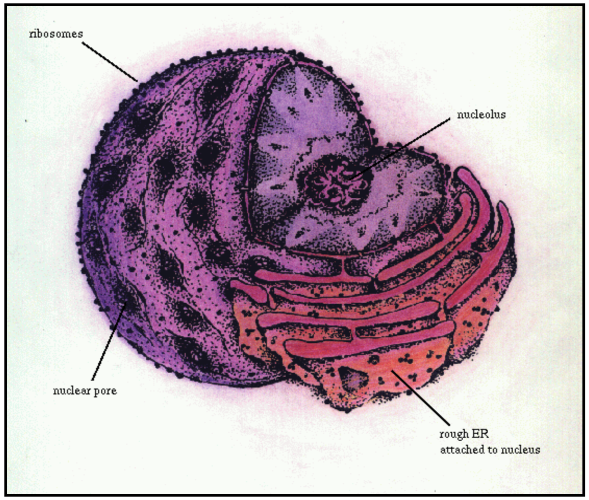
Endoplasmic Reticulum

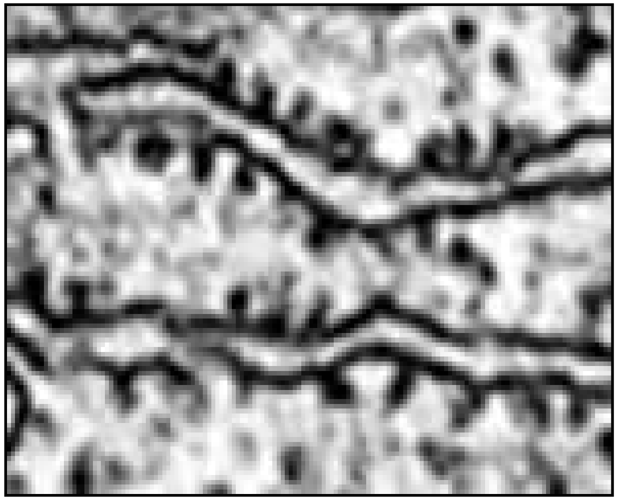


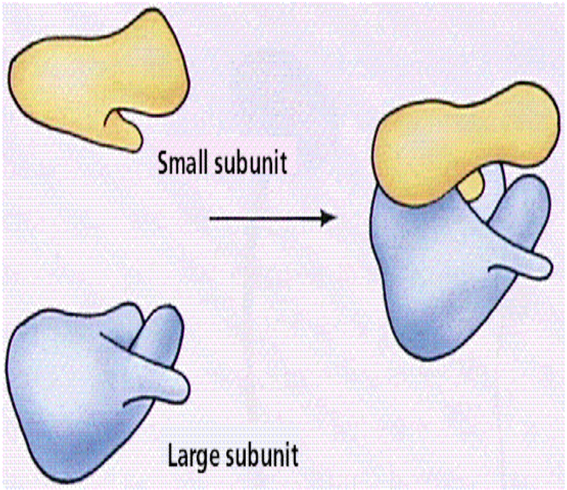


RIBOSOMES

- The sites of protein synthesis
- All living cells contain ribosomes, tiny organelles composed of approximately (**60%** RNA and **40%** protein).
- It can be described as protein factories



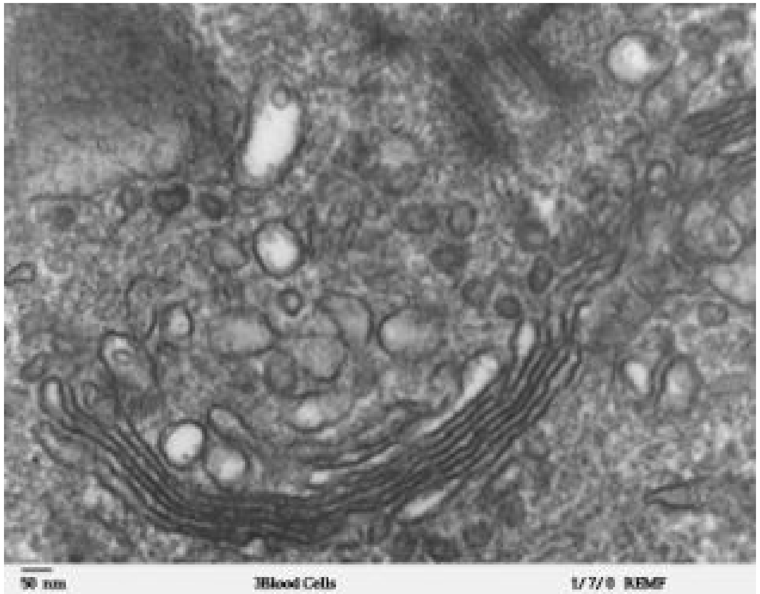


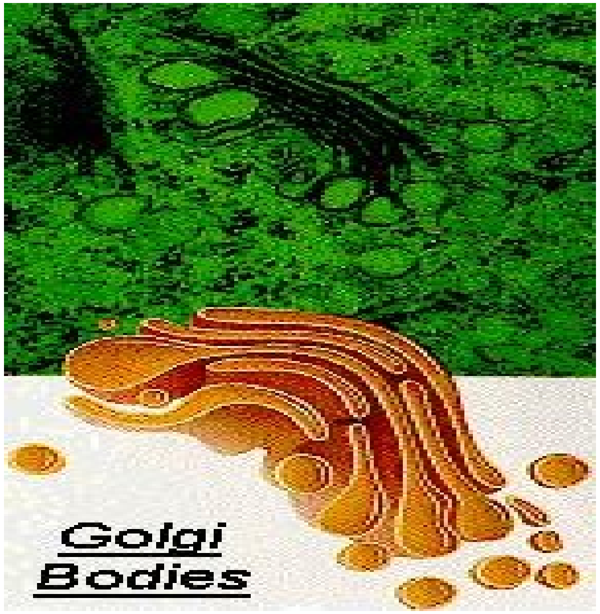


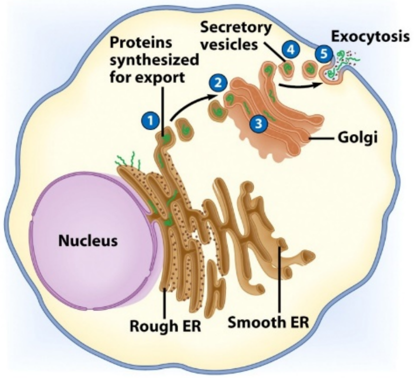
GOLGI APPARATUS

- Consists of series sacs
- The Golgi apparatus is the **distribution** and **shipping** department for the cell's chemical products.
- It **modifies** proteins and fats built in the endoplasmic reticulum and **prepares** them for export to the outside of the cell.

- Functions include **synthesis** of some substances, packaging of materials for transport and production of lysosomes .


50 nm
Blood Cells


Golgi
Bodies

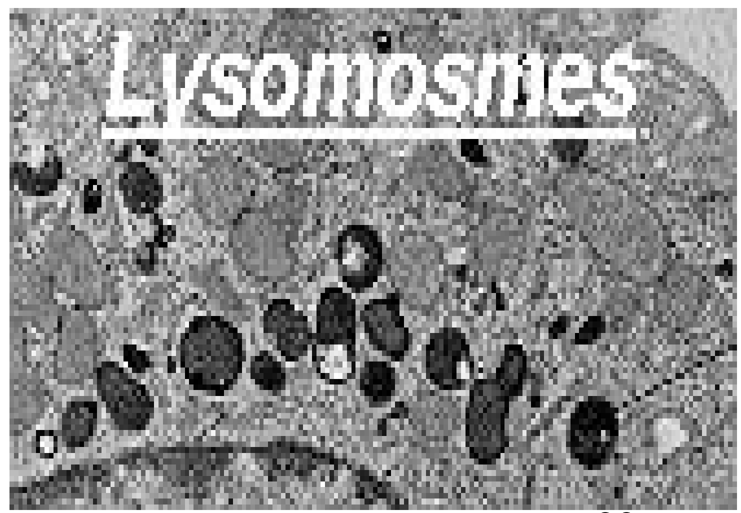

Secretory
vesicles
Exocytosis
Proteins
synthesized
for export
1
2
3
4
5
Golgi
Nucleus
Rough ER
Smooth ER

LYSOSOMES

- The main function of these Microbodies is **digestion**.

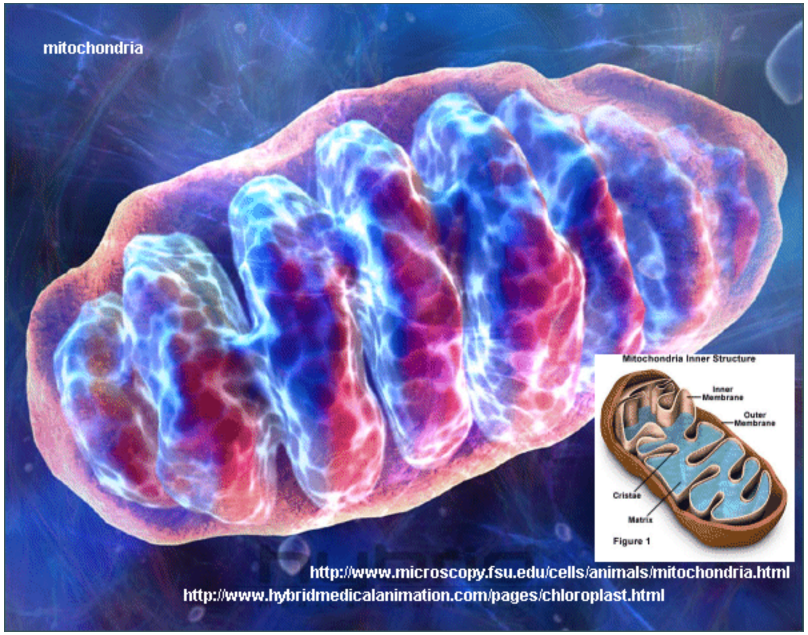
- Lysosomes **break down** cellular waste products and debris from outside the cell into simple compounds, which are transferred to the cytoplasm as new cell-building materials.

- Functions include **destruction** of damaged cells & digestion of strange materials (e.g. **bacteria**) .



Mitochondria

- Mitochondria are oblong shaped organelles that are found in the cytoplasm of every eukaryotic cell.
- In the animal cell, they are the main **power generators**, converting oxygen and nutrients into energy.
- Glucose+ oxygen=carbon dioxide+ water+ energy
- Contain their own **DNA**, **RNA** and **ribosomes**, and can **reproduce** themselves **independently** of the cell in which they are found.



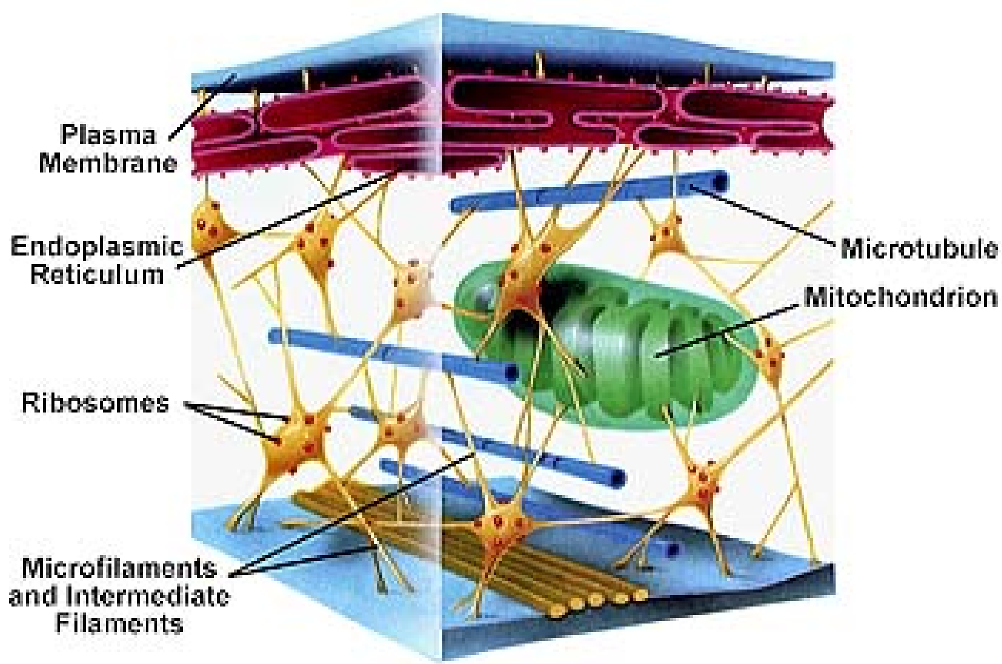
Mitochondria

PEROXISOMES

- Peroxisomes are the most common type of **Microbodies.**
- They are a diverse group of organelles that are found in the cytoplasm.
- Roughly spherical and bound by a single membrane.
- It carry out **Oxidation** reactions leading to production of **hydrogen peroxide** .

CYTOSKELETON

- Composed of **Microtubules**.
- Supports cell and provides shape.
- Aids movement of materials in and out of cells.



1. MICROTUBULES:

- It is one component of cytoskeleton
- Are straight, hollow cylinders are found throughout the cytoplasm of all eukaryotic cells (prokaryotes don't have them) and carry out a variety of functions; ranging from **transport** to **structural support**. (cytokinesis, vesicular transport).

2. MICROFILAMENTS:

- Are an important component of the cytoskeleton
- Are **solid rods** made of globular proteins called **actin**
- It is flexible ,functioning in **amoeboid movement** and **changes in cell shape.**

3. INTERMEDIATE FILAMENTS

- Intermediate filaments are important class of **fibrous proteins** that play an important role as both structural and functional elements of the cytoskeleton.

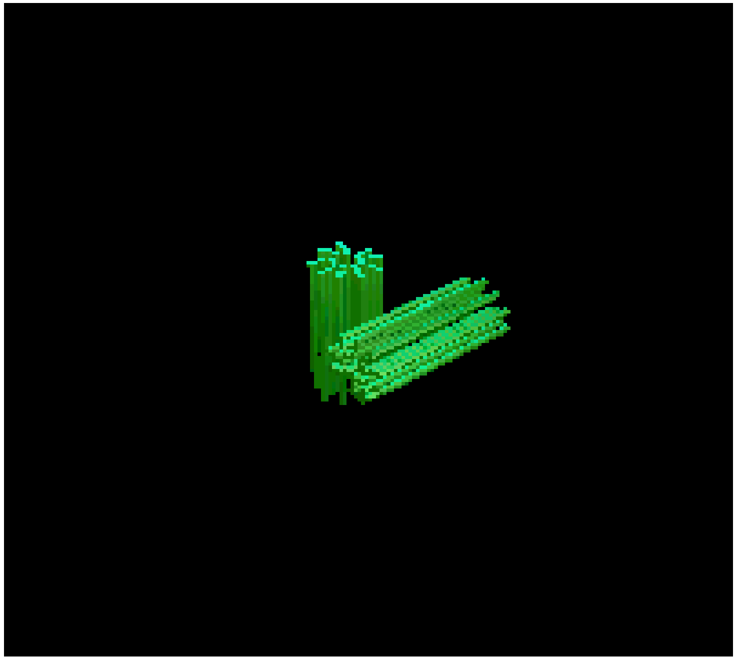
- Function to help maintain cell shape and rigidity.
- They may stabilize organelles, like nucleus

CENTRIOLES

- Centrioles are **self-replicating** organelles made up of **nine bundles of microtubules** and are found only in animal cells.

- They appear to help in organizing **cell division**.

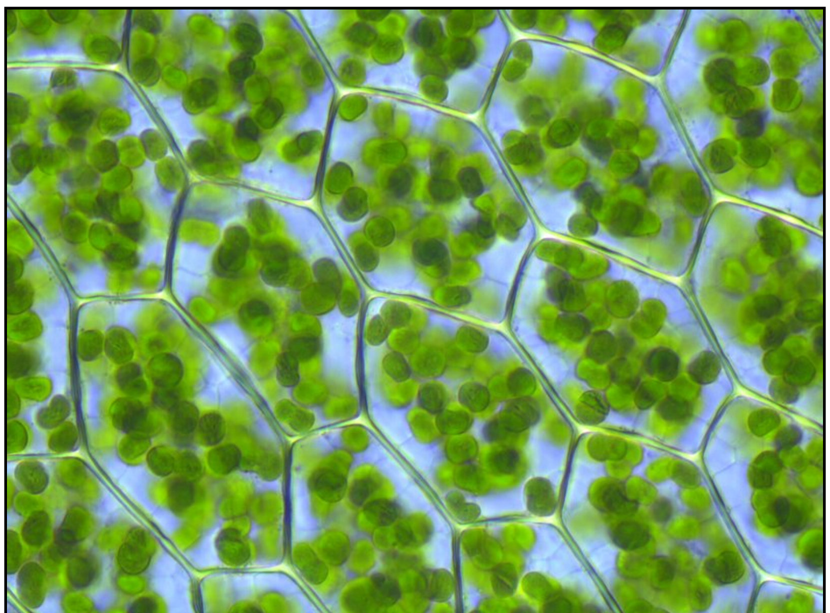
- It forms **spindle fibers** to
- separate chromosomes
- during cell division.

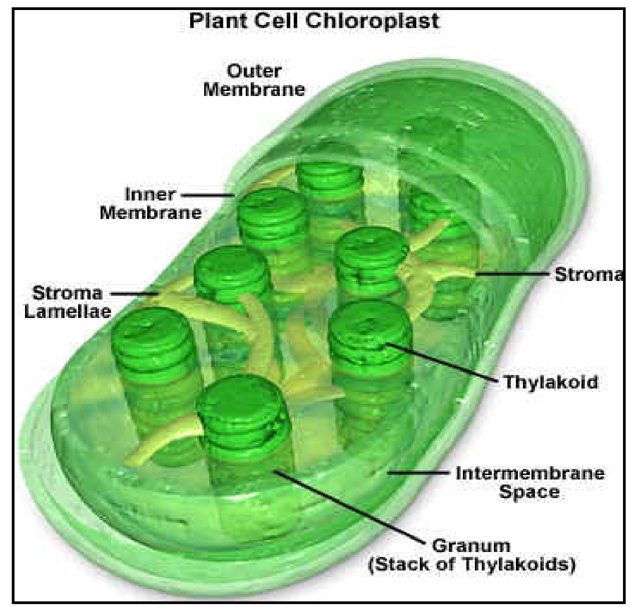


CHLOROPLAST

- Composed of double layer membrane
- It's the site of **photosynthesis**

(Carbon dioxide +water=glucose+ oxygen)

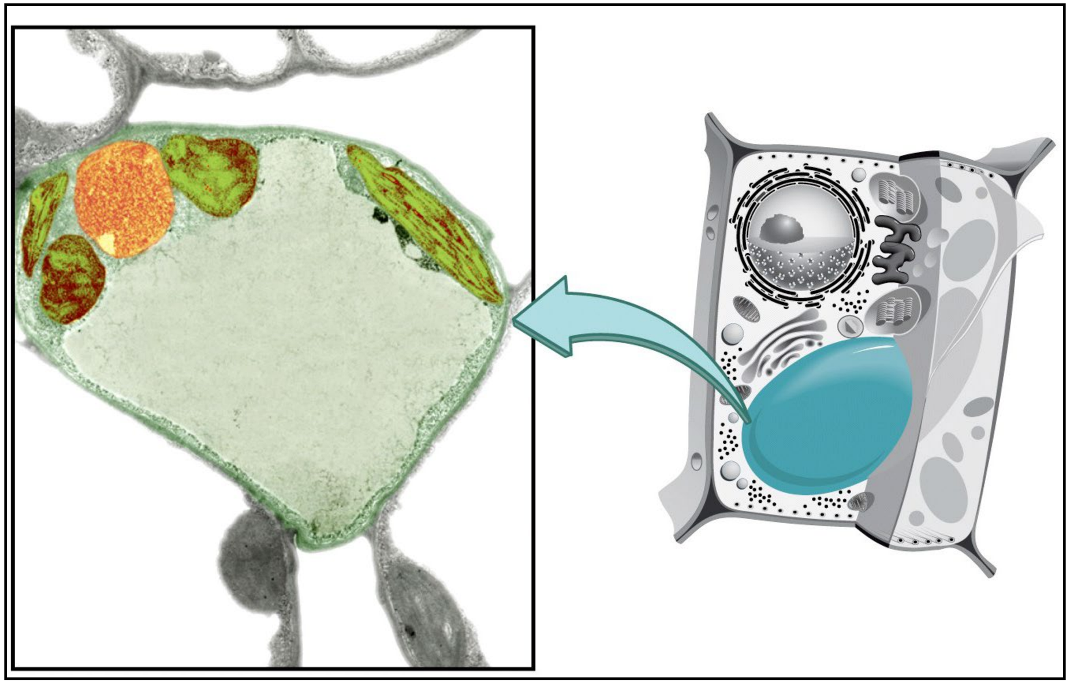




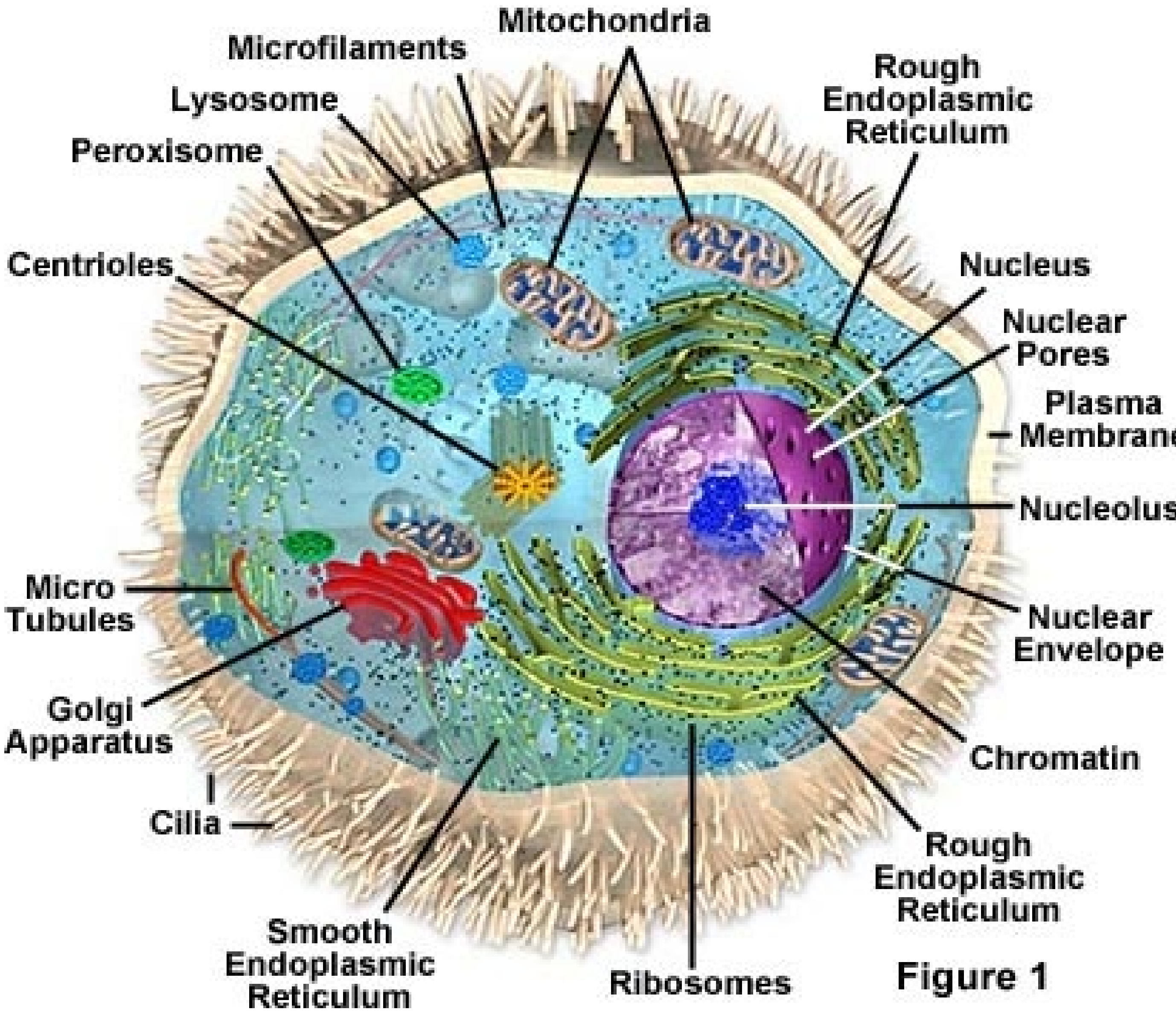
VACUOLE

- a single layer of unit membrane enclosing fluid in sac

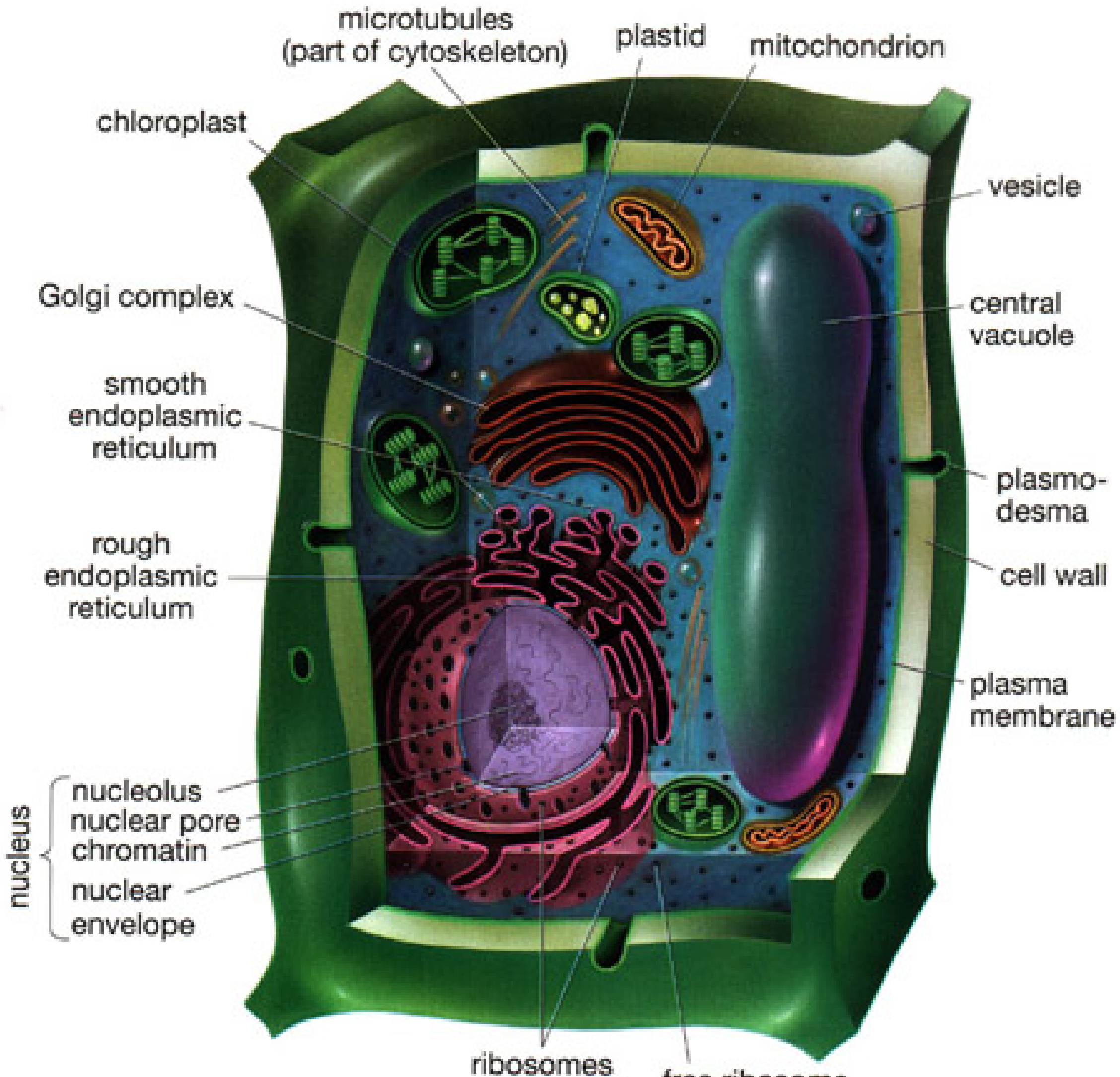
- Store **water** and **various chemicals** and help in **support** cell.



ANIMAL CELL


Mitochondria
Microfilaments
Lysosome
Peroxisome
Centrioles
Rough Endoplasmic Reticulum
Nucleus
Nuclear Pores
Plasma Membrane
Nucleolus
Micro Tubules
Golgi Apparatus
Cilia
Nuclear Envelope
Chromatin
Rough Endoplasmic Reticulum
Smooth Endoplasmic Reticulum
Ribosomes
Figure 1

PLANT CELL



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DIFFERENCES BETWEEN ANIMAL AND PLANT CELL

Plant cells	Animal cells
Possess rigid cell walls	No cell walls
Have green chloroplasts	No chloroplasts
Contain chlorophyll	Do not contain chlorophyll
Thin lining of cytoplasm	Most of cell is cytoplasm
Vacuole filled with cell sap	Small (if any) vacuoles

- **A cell resembles an independent country**:
- It has "borders": the plasma membrane which protect it
- It has a "central government" controlling its activity: the nucleus
- It has "transportation" systems: endoplasmic reticulum
- It has "energy" power-production factories: the mitochondria.
- It has "communication tools": RNA, receptor proteins
- It has "factories" manufacturing materials: Ribosomes
- It has "destruction weapons": Lysosomes

1- Nucleus

2- Chromosomes

3- Mitochondria

4- Ribosomes

5- Chloroplasts

6- Vacuoles

7- ER

8- Cell Membrane

THANK YOU